

SOFTWARE ENGINEERING LAB EXP-2

Name and ID

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Problem Statement –

Student Mental Health Monitoring System

SDGs Covered-

- SDG 3 – Good Health and Well-being
- SDG 4 – Quality Education

Overview of the Project

The Student Mental Health Monitoring System is an intelligent software-based platform designed to monitor and assess the mental well-being of students using data analytics. The system aims to identify early signs of mental health issues such as stress, anxiety, depression, and burnout that often go unnoticed in traditional support systems.

The system collects data from multiple sources including academic performance, attendance records, study patterns, sleep duration, screen time, behavioral indicators, and periodic self-assessment questionnaires. This data is analyzed to detect abnormal patterns or changes that may indicate mental distress.

Based on the analysis, students are classified into different risk levels, enabling counselors and authorized institutional staff to take timely and appropriate action. A centralized dashboard provides insights into individual and overall mental health trends while ensuring privacy, confidentiality, and ethical handling of sensitive student data.

By enabling early detection and intervention, the Student Mental Health Monitoring System promotes student well-being, improves academic outcomes, reduces dropout rates, and supports a healthier and more supportive educational environment, aligning with SDG 3 (Good Health and Well-being) and SDG 4 (Quality Education).

Use Case Diagram

Student Mental Health Monitoring System

Actors

1. Student
 2. Counselor / Mental Health Professional
 3. Admin (Institution Authority)
 4. System / Analytics Engine
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Student Use Cases

1. Register / Login
 2. Fill Self-Assessment Questionnaire
 3. View Mental Health Status
 4. View Personalized Feedback & Tips
 5. Track Mood / Well-being History
 6. Receive Notifications / Alerts
 7. Request Counseling Support
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Counselor Use Cases

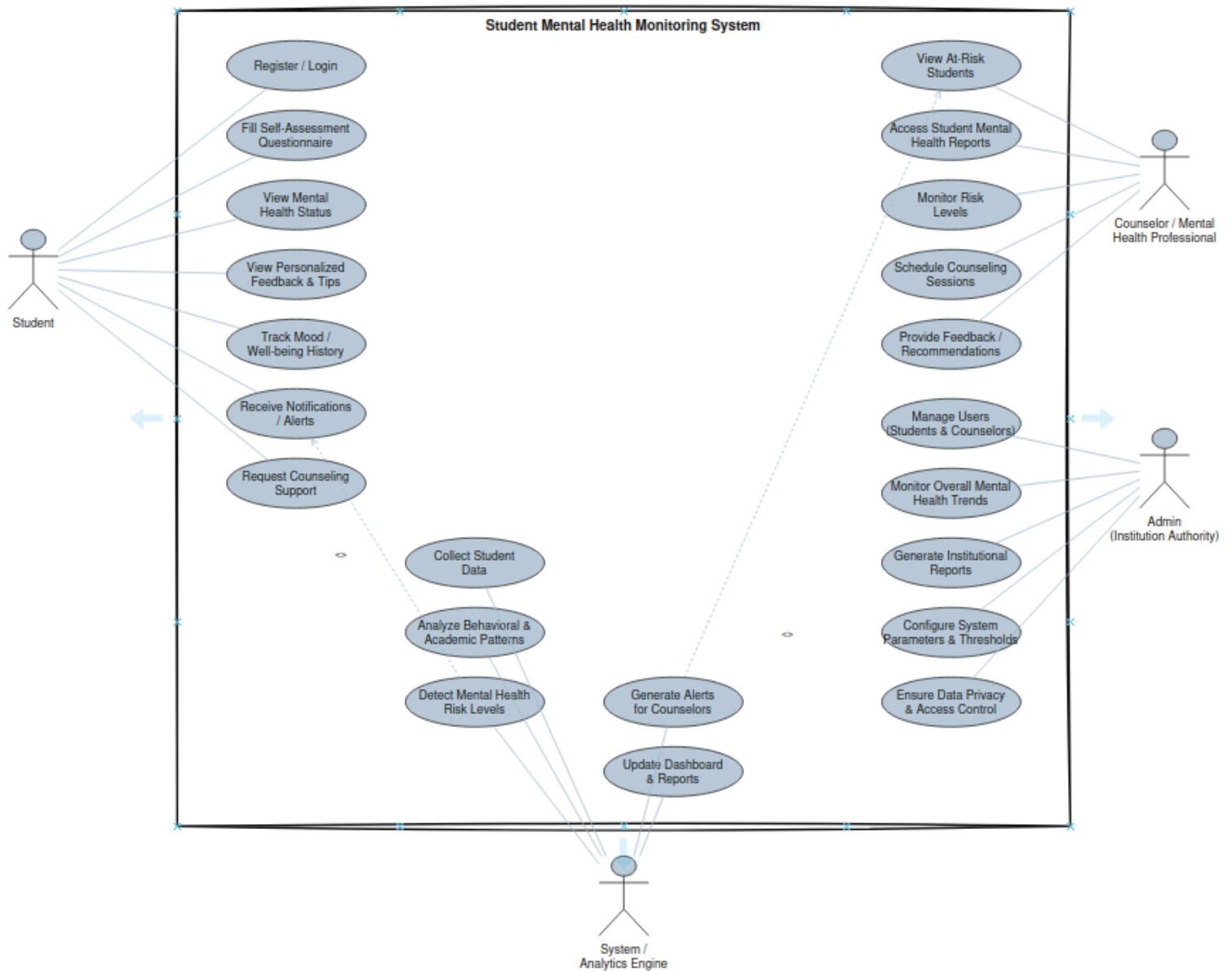
1. View At-Risk Students
 2. Access Student Mental Health Reports
 3. Monitor Risk Levels
 4. Schedule Counseling Sessions
 5. Provide Feedback / Recommendations
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Admin Use Cases

1. Manage Users (Students & Counselors)
 2. Monitor Overall Mental Health Trends
 3. Generate Institutional Reports
 4. Configure System Parameters & Thresholds
 5. Ensure Data Privacy & Access Control
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System / Analytics Engine Use Cases

1. Collect Student Data
2. Analyze Behavioral & Academic Patterns
3. Detect Mental Health Risk Levels
4. Generate Alerts for Counselors
5. Update Dashboard & Reports



LITERATURE SURVEY

1) Link: <https://digitalcommons.georgiasouthern.edu/gerjournal/vol19/iss1/4/>

Topic / Problem Statement: Many students experience emotional or behavioral difficulties that are not visible through academic performance alone, leading to missed opportunities for early intervention.

Approach Used:

- Universal mental health screening using the **Behavioral and Emotional Screening System (BESS)**.
- Screened all students, including those with mild symptoms.
- Emphasized brief, evidence-based tools easily implemented in schools.

Relation to Your Problem Statement:

- Addresses the lack of proactive monitoring by identifying at-risk students early.
- Reduces reliance on self-reporting or academic performance alone.
- Helps institutions provide early support, improving well-being and performance.

2) Link: <https://pubmed.ncbi.nlm.nih.gov/39314787/>

Topic / Problem Statement: Daily behavioral/emotional concerns can predict later mental health issues like anxiety and depression, yet routine observational data aren't systematically used.

Approach Used:

- Used **daily health observation scores** and **absenteeism trends** to monitor students.
- Correlated higher scores with later reports of depression/anxiety.

- Relied on structured observation rather than clinical assessments.

Relation to Your Problem Statement:

- Provides a **simple, structured monitoring framework** capturing early signs.
 - Useful where students hesitate to self-report due to stigma.
 - Supports data collection that can feed into intelligent early-warning systems.
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3) Link: <https://files.eric.ed.gov/fulltext/EJ1332345.pdf>

Topic / Problem Statement: Students with mild but emerging mental health symptoms are often missed, and relying on teacher observation alone is insufficient for early detection.

Approach Used:

- Suggested **standardized questionnaires** like PSC, SDQ, BESS for **universal screening**.
- Demonstrated the validity and reliability of systematic screening in schools.
- Included timely follow-up and referral plans.

Relation to Your Problem Statement:

- Strongly aligns with proactive, systematic detection rather than reactive counseling.
 - Supports institutional policies integrating screening into routine practice.
 - Helps identify students before academic or well-being decline.
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4) Link: <https://mental.jmir.org/2023/1/e42420>

Topic / Problem Statement: University students at risk for mental health issues are rarely identified until symptoms fully develop; structured annual health surveys may offer early cues.

Approach Used:

- Conducted **annual health surveys** with demographic and behavioral questions.
- Used patterns in responses — sometimes analyzed with machine learning — to flag risk.
- Focused on the predictive value of questionnaire design.

Relation to Your Problem Statement:

- Advocates **data-driven mental health monitoring** that doesn't depend on self-initiation.
 - Helps detect risk patterns before severe impact on academics or well-being.
 - Offers a scalable mechanism for institutional early detection.
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5) Link: <https://jurnal.globalhealthsciencegroup.com/index.php/IJGHR/article/view/4746>

Topic / Problem Statement: Bullying significantly increases the likelihood of anxiety, depression, and low self-esteem, but early warning signs often go unnoticed.

Approach Used:

- Scoping review of **questionnaire instruments** and **qualitative methods** for early detection among bullied adolescents.
- Emphasized teacher/counselor training to recognize warning signals.

- Linked early detection with supportive interventions.

Relation to Your Problem Statement:

- Highlights the importance of structured detection for high-risk groups.
 - Demonstrates how systematic frameworks catch distress that may be otherwise overlooked.
 - Reinforces the need for proactive systems rather than passive counseling.
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6)Link:

<https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.946998/full>

Topic / Problem Statement: Traditional questionnaires may not fully capture real-world student mental health challenges, reducing early detection accuracy.

Approach Used:

- Redesigned psychological screening instruments to better match real student profiles.
- Explored analysis with and without machine learning (e.g., decision tree models).
- Highlighted that improved screening design yields meaningful early indicators.

Relation to Your Problem Statement:

- Supports **intelligent, data-driven early detection systems** using well-constructed questionnaires.
- Shows that better tools improve early signs detection before major symptoms develop.
- Aligns with building systems that flag risk without waiting for self-reports.

